

Lithium borophosphate thin film electrolyte as an alternative to LiPON for solder-reflow processed lithium-ion microbatteries



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ABSTRACT

Microbatteries are energy sources well adapted to power various microsystems. To be considered as a microelectronic component, the microbatteries must be compatible with the solder-reflow treatment. During this industrial process, the temperature reaches 260 °C during few seconds. Having noticed an increase of the activation energy of the most used solid electrolyte lithium phosphate thin film (LiPON) after this type of thermal treatment, from 0.57 eV to 0.80 eV, we have investigated the correlations between composition, local structure (by XPS) and the electrical properties of lithium borophosphate solid electrolyte thin films. The lithium borophosphate thin films were prepared by radio-frequency sputtering using five different targets having the following compositions $x \text{LiBO}_2 \cdot (1-x) \text{Li}_3\text{PO}_4$ ($x = 5, 10, 15, 20$ and 25) in a pure argon discharge gas. In the absence of a thermal treatment, the addition of a small amount of boron (B/P ratio equal to 0.1) leads to an increase of the ionic conductivity ($1.1 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$) and a decrease of the activation energy (0.52 eV) compared to a pure lithium phosphate. However, for B/P ratio higher than 0.12, the ionic conductivity decreases. The study of the local structure by X-ray photoelectron spectroscopy (XPS) shows that the borate groups have less reacted with the phosphate groups leading to a phase separation reducing the connectivity between phosphate and borophosphate networks and so the ionic conductivity. The solder-reflow type thermal treatment leads to an increase of the ionic conductivity. The best ionic conductivity, $1.9 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$, is obtained for a B/P ratio equal to 0.24 with an activation energy of 0.57 eV instead of 0.80 eV for the LiPON thin film. For a use of Li-ion microbattery at low temperature after the solder-reflow type thermal treatment, the lithium borophosphate solid electrolyte would be a potential candidate to replace LiPON with several industrial advantages.

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1. Introduction

Ion-conducting glasses are an important class of solid electrolytes. Indeed, vitreous electrolytes have certain advantages over their crystalline counterparts like isotropic lithium ion migration and they allow a better control of their performances and properties with continuously variable compositions.

The microbattery is a monolithic system constituted by more than ten layers, including the active part (electrodes and solid electrolyte), protective layers, current collectors and thin film barriers. To be considered as a microelectronic component, the embedded microbattery needs to be compatible with the solder-reflow process, which reaches a temperature of 260 °C during few tens of seconds, and to have the same lifetime during utilization and storage than for other electronic components. So to be integrated in a Li-ion microbattery, the solid

electrolyte thin film has to survive the manufacturing process and the various thermal treatments.

Lithium phosphorus oxynitride amorphous glass thin film (LiPON) is widely used in solid-state microbatteries as a solid electrolyte [1]. Bates et al. were the first to report on the preparation of the so-called LiPON electrolyte by reactive radio-frequency sputtering from a Li_3PO_4 target in pure nitrogen discharge gas [2,3]. Several percents of nitrogen atoms were incorporated into the thin films in place of oxygen atoms. This material exhibits a good chemical stability, a quite high ionic conductivity, close to $3.0 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$ at room temperature as well as a very low electronic conductivity. Numerous studies dealing with the preparation of LiPON thin films are reported in the literature showing the existence of numerous types of $\text{Li}_x\text{PO}_y\text{N}_z$ with compositions ranging from 2.6 to 3.5 for x , from 1.9 to 3.8 for y and between 0.1 and 1.3 for z [3–6]. Various deposition parameters were investigated including radio-frequency power, N_2 pressure, target-substrate distance. In these studies, the ionic conductivity increases with the nitrogen content [3–7]. In the same time, the activation energy decreases. In our previous studies [8], we have clearly demonstrated that the improvement of the ionic conductivity combined with the decrease of activation energy

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stems from an increase of the lithium ion mobility. It was also highlighted in previous study [9] that the solder-reflow type thermal treatment at 260 °C leads to an improvement of the ionic conductivity from $3.0 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$ to $7.2 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$ at room temperature as well as an increase of the activation energy from 0.57 eV to 0.80 eV, by the formation of Li_2O and $\text{Li}-\text{O}-\text{P}$ type environments, which is detrimental for a use of Li-ion microbatteries at low temperature. So, there is a real need to improve or at least to maintain the electrolyte performances after solder-reflow in order to allow a use of the microbattery below room temperature.

The substitution of phosphorus oxide by boron oxide for a constant alkali content can have a significant influence on the glass properties. Carta et al.; [10] had shown that the addition of B_2O_3 in sodium phosphate glass network decreases the hygroscopy of these glasses. The works of Zielenok et al. [11,12] confirm this result and demonstrate that the addition of B_2O_3 to a phosphate network improves the ionic conductivity, the chemical durability as well as the thermal and mechanical stability compared to pure phosphate glasses. The same results were observed previously by Hudgens et al. [13]. These evolutions were explained by an increase of the bond strength and a cross-linking of the network by formation of $\text{P}-\text{O}-\text{B}$ bonds, as demonstrated by Cho et al. [14], Qiu et al. [15] and Raskar et al. [16]. Nevertheless, as observed by Money et al. [17], for the lithium borophosphate bulk glasses, the increase of boron oxide content, beyond 20% mol, may cause micro-phase separation leading to boron-rich domains and a decrease of the ionic conductivity with an increase of the activation energy.

It has to be noted that the relationship between the local structure, the properties and the thermal treatment of lithium borophosphate thin films have drawn very little attention in comparison to lithium phosphate bulk glasses. To our knowledge, the effect of the solder-reflow on the lithium borophosphate thin films has never been reported in literature, whereas this point is crucial for the development of microbatteries.

This study deals with the preparation by radio-frequency sputtering and the characterization of lithium borophosphate glasses thin films. The latter were sputtered using different targets having the following compositions $x \text{ LiBO}_2$ ($100 - x$) Li_3PO_4 ($x = 5, 10, 15, 20, 25$). Impedance spectroscopy was used to determine the evolution of electrical performances as ionic conductivity and activation energy. X-ray photoelectron spectroscopy was used to investigate the local structure. Some correlations were established between the chemical composition, the local structure and the ionic conductivity. Finally, the first promising electrical performances obtained after solder-reflow will be presented.

2. Experimental procedure

2.1. Sample preparation

Lithium borophosphate (LiPOB) thin films were deposited by radio-frequency (rf) magnetron sputtering, with argon as discharge gas, in a chamber (PLASSYS) connected to an argon-filled glove box. We decided to limit the boron molar percentage at 25% to avoid the thin film crystallization reported by Money et al. [17]. Hence, the 50 mm diameter target was prepared by uniaxial cold pressing mixture of $\beta\text{-Li}_3\text{PO}_4$ (Aldrich, 99.9%) and LiBO_2 (Aldrich, 99.9%) powders considering the five following compositions: LiBO_2 : ($100 - x$) Li_3PO_4 with $x = 5, 10, 15, 20$, and 25. Due to the polymorphic phase transformation from $\beta\text{-Li}_3\text{PO}_4$ to polymorphic $\gamma\text{-Li}_3\text{PO}_4$ at 520 °C, Kim et al. [18] have observed the formation of many cracks in the target. Therefore, the pellet was first annealed at 850 °C for 12 h to prevent any further cracking. After this thermal treatment, the powder was crushed, pressed again and the pellet annealed at 600 °C for 12 h. The target density is close to $1.9 \text{ g} \cdot \text{cm}^{-3}$ which corresponds to 80% of the theoretical density. XRD analysis shows that all targets are a mixture of $\gamma\text{-Li}_3\text{PO}_4$ phase (orthorhombic system, Pmnb) and $\beta\text{-LiBO}_2$ phase (monoclinic system, P2₁/c). Before each deposition, a vacuum was created in the chamber until the

pressure was less than 4.10^{-4} Pa and a pre-sputtering was systematically applied during 1 h. All thin films were deposited at room temperature with no intentional heating of the substrates, with a nominal radio-frequency power density of $2 \text{ W} \cdot \text{cm}^{-2}$ and a target-to-substrate distance equal to 8 cm. The sputtering was performed with a total pressure of 1 Pa for five and a half hours under a pure argon atmosphere, the argon flow rate being equal to $40 \text{ ml} \cdot \text{min}^{-1}$. Various substrates were used according to the envisaged characterization. In order to avoid any surface contamination of the films, all samples were handled in an argon-filled glove box and transferred to the different apparatus used for characterization through airtight containers.

2.2. Characterization techniques

The thickness of LiPOB thin films deposited on Si substrate was determined using a Tencor profilometer alpha-step 200. The thickness mean is around 1.5 μm .

The P, O and B concentrations were determined on thin films deposited on Si substrate by electron microprobe analysis (Cameca SX100) using InP, $\text{Y}_3\text{Fe}_5\text{O}_{12}$, B and Si (for substrate) as calibration standards. The Li, P and B contents were measured using an inductively coupled plasma-optical emission spectrometer (ICP-OES) (Varian 720ES) by dissolving beforehand thin films deposited on glass substrate in 10 ml of HCl. A small quantity of the solutions was injected. The wavelength of the emission line is 460.289 nm for Li, 214.914 nm for P and 249.678 nm for B.

The ionic conductivity of LiPOB thin films was measured by impedance spectroscopy using a Solartron 1260 analyzer. Stainless steel/electrolyte/stainless steel sandwich structures were prepared on glass substrates. A special design with a grid allowing 4 measurements for each thin film electrolyte was used to ensure the reproducibility and a better accuracy of the results. The overlap area of each plot was 4 mm^2 (Fig. 1). The stainless steel electrodes were deposited by radio-frequency magnetron sputtering from a stainless steel target under Ar atmosphere. The impedance measurements were carried out in a frequency range from 1 Hz to 10 MHz with 50 mV ac voltage amplitude with a 5 °C increment from 30 to 125 °C. The ionic conductivity, σ , was calculated from the following equation:

$\sigma = \frac{1}{R} \times \frac{d}{A}$ where d is the solid electrolyte thin film thickness, A the area of the electrode and R the solid electrolyte thin film resistance deduced from the Nyquist diagram. The activation energy was obtained by the study of the ionic conductivity temperature dependence. This dependence follows Arrhenius equation and the straight obtained has only one coefficient factor over the temperature range studied.

X-ray photoelectron spectroscopy (XPS) was used to investigate the local structure of thin films. XPS analyses were carried out with a VG Escalab 220i XL spectrometer using a focused and monochromatized Al K α radiation (1486.7 eV) (diameter of the irradiated area = 150 μm) under a residual pressure of $8 \times 10^{-9} \text{ mbar}$ on thin films deposited on a silicon substrate. The spectrometer was calibrated using the photo-emission lines of gold ($\text{Au } 4f_{7/2} = 83.9 \text{ eV}$, with reference to the Fermi

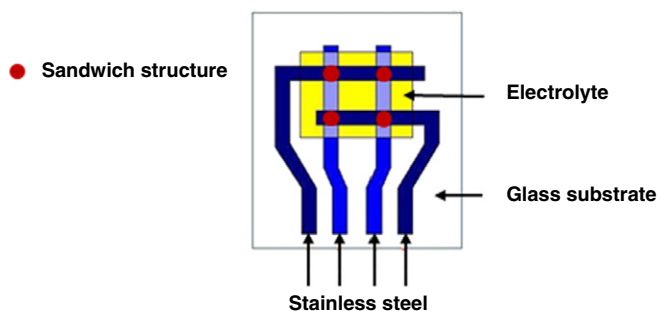


Fig. 1. A schematic view of stainless steel/electrolyte/stainless steel (sandwich structure) used for ionic conductivity measurements.

level) and copper (Cu 2p_{3/2} = 932.5 eV). The Au 4f_{7/2} full width at a half maximum (FWHM) was 0.86 eV. In this work, the peaks were recorded with constant pass energy of 30 eV. All the samples were fixed on sample holders in a glove box directly connected to the introduction chamber of the spectrometer. The XPS analyses were carried out on the as-deposited thin film. Peaks were then shifted to align hydrocarbon C1s photoemission to 285.0 eV binding energy. The experimental curves were fitted with a combination of Gaussian (80%) and Lorentzian (20%) functions using a minimum number of components.

The solder-reflow type thermal treatment was done in a glass tube under primary vacuum to avoid any surface contamination. This type of thermal treatment was choice because, in the industrial point of view, the solder-reflow type thermal treatment will be occurred after the end of fabrication process, so the solid electrolyte will never see the air atmosphere. Moreover, to analyze very well the local structure of solid electrolyte by XPS, we must avoid the air, carbon and oxygen contamination which would increase during the air atmosphere thermal treatment. The heating rate was 30 °C·s⁻¹ to reach 260 °C and the sample was maintained at this temperature during 2 min to insure a homogeneous temperature. The cooling was obtained with the ambient environment. We tried to reproduce the most faithfully the lead-free alloy reflow profile with this procedure.

3. Results and discussions

3.1. Before the solder-reflow type thermal treatment

In what follows, to facilitate understanding, the samples are named as follows: LiPOB-x, where x corresponds to the molar percentage of lithium borate in the relative target from which the sample was prepared. To have a reference compound, a thin film was deposited by rf sputtering under argon discharge gas from a β -Li₃PO₄ target called LiPOB-0.

The values of the chemical composition deduced from microprobe analysis and ICP are collected in Table 1. As expected, when the boron content in the target increases, the B/P ratio and so the boron content incorporated in the thin film also increases. We could expect that the lithium and oxygen contents in thin film decrease with the increase of LiBO₂ content in the target, because a part of Li₃PO₄ is replaced by a compound less rich in lithium and oxygen. Nevertheless, we observe the opposite. This can be explained by the fact that lithium, boron and oxygen elements are lighter than the phosphorus, they are more easily sputtered under argon discharge gas. The chemical composition of the different targets after deposition process confirms these results. The targets were depleted of lighter elements and so were not sputter stoichiometrically. Furthermore, the boron content in the thin film is consistently higher than in the relative target, expect for the boron richest content. Hence, whatever the composition of the starting target, the Li/(B + P) and O/(B + P) ratios remain constant in the different thin layers. Thus, only the boron and phosphorus amounts vary, which allowed us to study “the eventual mixed-former effect” on the properties of our thin films.

Table 1
Chemical compositions of the different LiPOB-type thin films determined by ICP and EPMA.

Samples	Li/P (±0.02)	O/P (±0.05)	B/P (±0.02)	Chemical composition
LiPOB-0	3.83	4.35	0.00	Li _{3.83} PO _{4.35}
LiPOB-5	3.00	3.99	0.09	Li _{3.00} PO _{3.99} B _{0.09}
LiPOB-10	3.10	4.06	0.14	Li _{3.10} PO _{4.06} B _{0.14}
LiPOB-15	3.29	4.32	0.18	Li _{3.29} PO _{4.32} B _{0.18}
LiPOB-20	3.52	4.58	0.22	Li _{3.52} PO _{4.58} B _{0.22}
LiPOB-25	3.59	4.64	0.24	Li _{3.59} PO _{4.64} B _{0.24}

The ionic conductivity was deduced from impedance measurements. An example of impedance spectrum recorded at 25 °C for LiPOB-5 thin film is given in Fig. 2a. All spectra obtained for the different samples are similar. The solid line superimposing the data points corresponds to the calculated values of the impedance using an equivalent circuit. In the equivalent circuit used to fit the experimental curves (Fig. 2b), R_{cc} represents the resistance of current electrodes and Z_i the impedance of both the top and bottom electrode-electrolyte junctions. The bulk response is represented by Z_{el} element in parallel with R_{el}. Z_i and Z_{el} are both constant phase elements (CPE). To have a good fit, it was not necessary to consider an interface between the stainless steel electrodes and the solid electrolyte. The activation energy was obtained by the study of the ionic conductivity temperature dependence. This dependence follows Arrhenius equation and the straight obtained has only one coefficient factor over the temperature range studied, from 25 °C to 120 °C. The evolutions of the ionic conductivity at 25 °C (room temperature) and the activation energy as a function of the B/P ratio are shown in Fig. 3. At room temperature, the ionic conductivity of the LiPOB-5 thin film is greatly enhanced compared to pure lithium phosphate thin layer deposited under argon discharge gas (LiPOB-0), from 7.0 × 10⁻⁷ S·cm⁻¹ to 1.1 × 10⁻⁶ S·cm⁻¹, while the activation energy decreases from 0.64 eV to 0.52 eV. This improvement could be attributed to the “mixed-former effect” in agreement with previous published studies presented in Introduction section on lithium or sodium borophosphate bulk glasses [10,11,15–17,19]. Surprisingly, for a higher boron content (B/P > 0.1) and contrary to studies achieved on bulk materials, the ionic conductivity decreases with the addition of boron, reaching a constant value around 8.0 × 10⁻⁷ S·cm⁻¹ for B/P > 0.16 while the activation energy increases to about 0.57 eV. However, whatever boron content, the ionic conductivity of the thin film is systematically higher than for the pure phosphate one.

Thus, it seems that the “mixed-former effect” could be observed for a lower boron content in our lithium borophosphate thin films compared to the corresponding bulk glasses, as especially presented by Zielonok et al. [11,12].

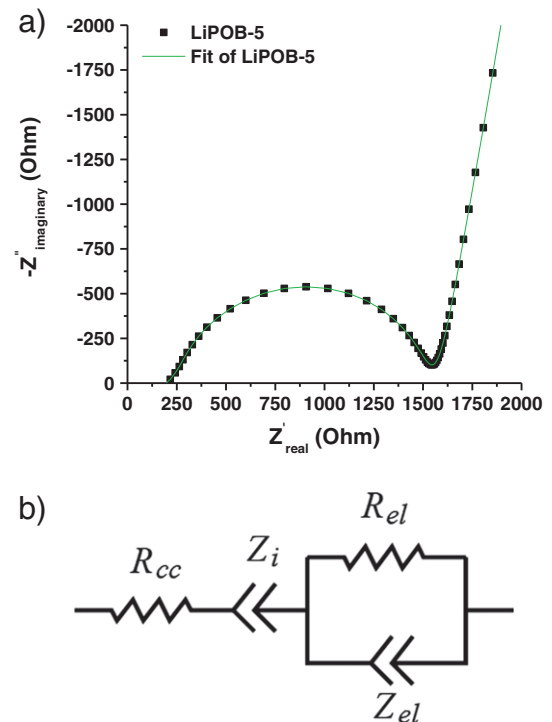


Fig. 2. a) Impedance spectra of LiPOB-5 thin film and fit with the equivalent circuit before the solder-reflow thermal treatment, recorded at 25 °C. b) The equivalent circuit used to fit the impedance spectra.

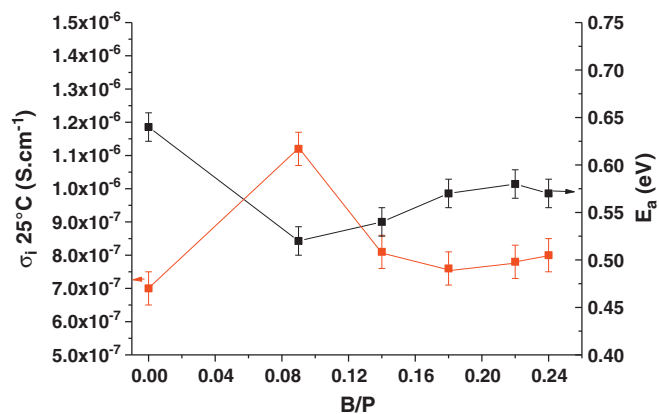


Fig. 3. Ionic conductivity σ_i in red square recorded at 25 °C and activation energy E_a in black triangle as a function of B/P ratio in the thin film (without solder-reflow thermal treatment).

To better understand these evolutions with the boron content and the relationship with the eventual “mixed-former effect”, all these thin films were characterized by X-ray photoelectron spectroscopy on the basis of core peak analysis. First, it has to be noted that the surface chemical composition of the different films, obtained by X-ray photoelectron spectroscopy, is in agreement with the chemical composition determined by electron microprobe analyses and ICP, these two last techniques allowing an in-depth analysis of the samples. For all samples, the carbon photoelectron peak C1s displays only one component located at 285.0 eV attributed to carbon contamination C–C always present at materials surface (around 10 at.%). So all the oxygen components detected by XPS are relative to the material and not the surface contamination, confirming the admissibility of our thermal treatment process. For all samples, the XPS spectra of P2p display two resolved doublets (with a spin-orbit splitting of about 0.9–1.0 eV between $2p_{3/2}$ and $2p_{1/2}$): the first doublet located at 133.4–134.4 eV is characteristic of a PO_4^{3-} tetrahedral environment as in $\gamma\text{-Li}_3\text{PO}_4$ phase or glass [20] while the second one at 134.2–135.2 eV reveals a P–O–P environment [8]. As only oxygen atoms are connected to the phosphorus atoms, no change in binding energy can be observed. Only the variation of phosphorus atomic ratio in the material can be determined. For all samples, the XPS spectra of B1s display only one component located at 191.6 eV which can be attributed to B–O bonds [19]. As for phosphorus atoms, only oxygen atoms are connected to the boron atoms. So no change in binding energy can be observed and only variation of boron atomic ratio in the thin film can be determined. Thus, only the analysis of the oxygen photoelectron peak allows the discrimination of bridging and non-bridging oxygen atoms in the borate or phosphate glasses. The XPS O1s core peak spectra for LiPOB thin films are shown in Fig. 4 and the relative data are presented in Table 2. The XPS experimental curves were fitted with a combination of Gaussian and Lorentzian functions using a minimum number of components and the full width at half maximum (FWHM) of these components is identical for a same core peak.

For pure phosphate glasses, the O1s spectrum was decomposed into three components respectively attributed to $\text{P}=\text{O}$ (component O_1 : non-bridging oxygen at 531.5 eV), $\text{Li}^+\cdots\text{O}-\text{P}$ (component O_2 : non-bridging oxygen at 532.4 eV) and $\text{P}-\text{O}-\text{P}$ (component O_3 : bridging oxygen at 533.6 eV) based on structural model presented in Fig. 5. This O1s XPS spectrum is similar to those previously reported for alkali metaphosphate bulk glasses [21,22] and our previous results [8] and the fit according to the work of Brückner et al. [23] and Marchand et al. [21]. When the boron content increases, the break of experimental curve slope around 533.0 eV tends to decrease, suggesting the presence of a fourth component O_4 which increases gradually with the rate of boron. This new component located at 533.0 eV can be attributed to

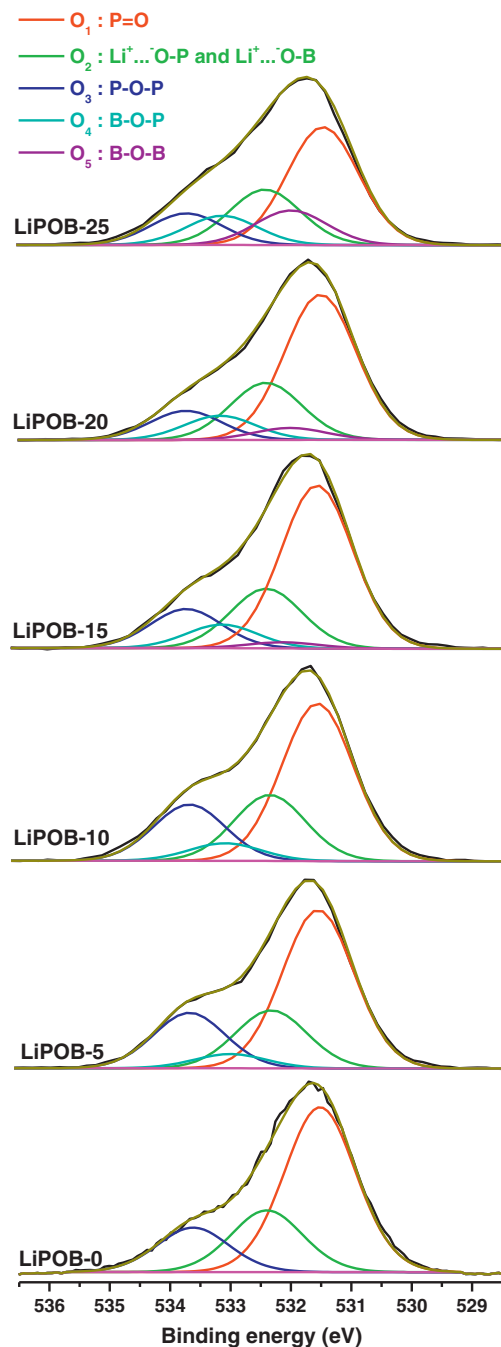


Fig. 4. O1s XPS spectra of the as-deposited LiPOB-x thin films.

an oxygen atom bound to one boron atom and one phosphorus atom B–O–P, according to the studies of Raskar et al. [16]. This deconvolution with only four components does not allow to obtain an agreement with the experimental curve, especially for the O1s core peak level of LiPOB-15, -20 and -25 between 531.0 eV and 533.0 eV, because of the increase of the full width at half maximum of the experimental curve in this binding energy region. By keeping the same binding energies for the O_1 , O_2 , O_3 and O_4 components and the same FWHM for all components, we consider a fifth component for these three spectra located at 532.0 eV, characteristic of an oxygen atom bound to two boron atoms B–O–B in agreement with the study of Raskar et al. [16]. Fig. 6 shows the evolution of the relative proportions of the different types of bridging oxygen as a function of the boron content.

Table 2

XPS O1s binding energies (eV) of the five components for as-deposited LiPOB thin films. The relative percentage of the different O1s components is indicated as well as the full width at half maximum values in parentheses. The last column indicates the total oxygen atomic percentage.

Sample	B.E (eV) – FWHM (eV) – %	At. %
LiPOB-0	O ₁ 531.5 – (1.4) – 61	27.1
	O ₂ 532.4 – (1.4) – 24	10.7
	O ₃ 533.6 – (1.4) – 15	6.7
		44.5
LiPOB-5	O ₁ 531.5 – (1.4) – 55	25.2
	O ₂ 532.3 – (1.4) – 20	9.2
	O ₃ 533.7 – (1.4) – 20	9.2
	O ₄ 533.0 – (1.4) – 5	2.3
		45.9
LiPOB-10	O ₁ 531.6 – (1.4) – 53	23.8
	O ₂ 532.4 – (1.4) – 22	9.9
	O ₃ 533.7 – (1.4) – 19	8.9
	O ₄ 533.1 – (1.4) – 6	2.7
		45.3
LiPOB-15	O ₁ 531.5 – (1.4) – 52	23.6
	O ₂ 532.4 – (1.4) – 21	9.6
	O ₃ 533.7 – (1.4) – 16	7.3
	O ₄ 533.1 – (1.4) – 8	3.6
	O ₅ 532.1 – (1.4) – 3	1.4
		45.5
LiPOB-20	O ₁ 531.5 – (1.4) – 50	23.2
	O ₂ 532.4 – (1.4) – 21	9.8
	O ₃ 533.7 – (1.4) – 14	6.5
	O ₄ 533.2 – (1.4) – 9	4.2
	O ₅ 532.0 – (1.4) – 6	2.8
		46.5
LiPOB-25	O ₁ 531.5 – (1.4) – 44	20.5
	O ₂ 532.4 – (1.4) – 21	9.8
	O ₃ 533.7 – (1.4) – 12	5.6
	O ₄ 533.1 – (1.4) – 11	5.1
	O ₅ 532.0 – (1.4) – 13	6.0
		47.0

For all samples, the Li1s spectra display only one narrow peak at 55.7 eV which can be assigned to lithium atoms in an environment like $\text{Li}^+ \cdots \text{O}-\text{P}$ and/or $\text{Li}^+ \cdots \text{O}-\text{B}$ [8,9]. Indeed, the electronegativity difference between boron and phosphorus atoms is not high enough to have a modification of the binding energy of lithium atoms. So for all samples with boron, the component O₂ located at 532.4 eV can be attributed to non-bridging oxygen $\text{Li}^+ \cdots \text{O}-\text{P}$ and $\text{Li}^+ \cdots \text{O}-\text{B}$. Whatever the sample, the relative proportion of the non-bridging oxygen $\text{Li}^+ \cdots \text{O}-\text{P}$ and $\text{Li}^+ \cdots \text{O}-\text{B}$ (component O₂) remains constant at around 10 at.%. This result is consistent with the constant rate of lithium (Li/(B + P) ratio is constant) obtained by electron microprobe and ICP.

With a low addition of boron, from LiPOB-0 to LiPOB-5, the relative proportion of the O₁ component (P=O) decreases whereas one of the O₃ (P–O–P) and O₄ (B–O–P) components increase. This evolution would be consistent with the formation of a partial hybrid

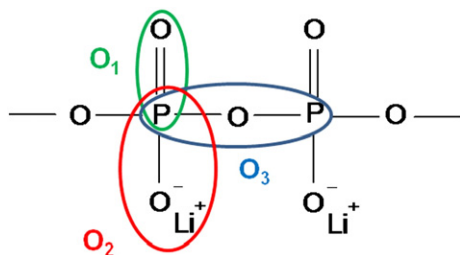


Fig. 5. Structural model of alkali metaphosphate glasses with attribution of O1s components.

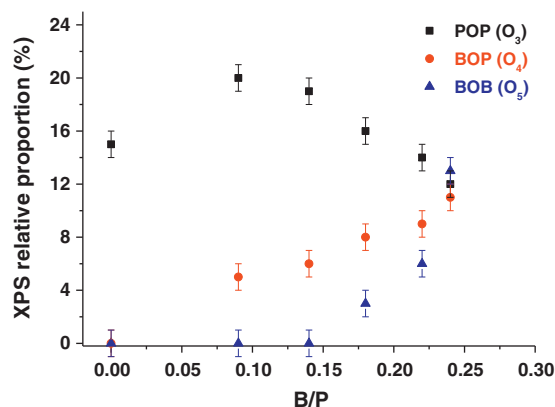


Fig. 6. Evolution of O₃ (P–O–P), O₄ (B–O–P) and O₅ (B–O–B) relative proportions deduced from their respective areas on the O1s core peak.

borophosphate network formed by the reaction of all BO₃ units with a part of PO₄ units leading to tetrahedral environment for B and P (Fig. 7). The term of partial hybrid borophosphate network is used because only a part of phosphate reacts with all borate entities to create a mixture of borophosphate network and phosphate network. The mobility of the lithium ions was so enhanced by the relocation of the negative charge on the borate units (BO₄), the phosphate units connected to these borate units being neutral in the borophosphate network. This induces the improvement of the ionic conductivity and the decrease of the activation energy. In the same time, the proportion of oxygen atoms in an environment like P–O–P increases, reflecting a cross-linking of the glass network.

From LiPOB-5 to LiPOB-20, the increase of boron content leads to a decrease of the proportion of O₃ component (P–O–P environment) from 9.2 at.% to 6.5 at.%, in agreement with the decrease of phosphorus content, while the O₄ (P–O–B environment) and O₅ components (B–O–B environment) increase, respectively from 2.3 at.% to 4.2 at.% and from 0 at.% to 2.8 at.%. The proportion of oxygen atoms in an environment like B–O–B strongly increases for the LiPOB-25 (around 6.0 at.% instead of 2.8 at.% for the LiPOB-20) meaning that a great number of borate units are linked together. The decrease of the O₁ component (P=O bonds) could be correlated with the decrease of phosphorus content but also with the formation of B–O–P bonds (Fig. 6). The evolution of the local structure of the lithium borophosphate thin films is quite different from the one of the corresponding bulk glasses [13,16,17]. The proportion of B–O–P bonds is lower in our thin films while the number of B–O–B bonds is much more important. Contrary to the LiPOB-5 sample whom all the borate units reacted with a part of phosphate units to form a hybrid borophosphate network, we can expect a new network formation, which would be a boron-rich neutral network. So in fact, we could have a mixture of three networks: a hybrid borophosphate network in agreement with the proportion of B–O–P bonds, a boron-rich neutral network in agreement with the presence of B–O–B environment and a phosphate network. As the borate units react less with the phosphate units in comparison with the LiPOB-5 sample, the new boron rich network prevents the conversion of BO₃ into BO₄ and decreases the cross-linking in the glass network which leads to a decrease of the charge carrier mobility at the origin of the ionic conductivity decrease.

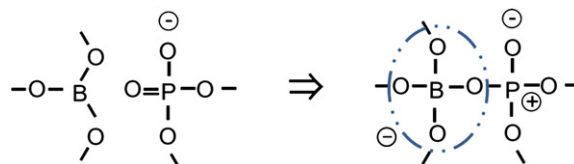


Fig. 7. The formation of the hybrid phosphate network.

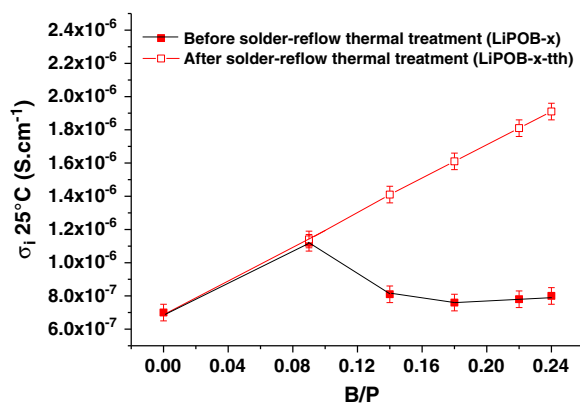


Fig. 8. Ionic conductivity recorded at 25 °C before and after solder-reflow thermal treatment as a function of B/P ratio.

To sum up, when the boron content is low (typically $B/P < 0.12$), the glass network tends to maintain the local structure characteristic of the phosphate network with additional formation of borophosphate units which increase the network cross-linking. The delocalization of the negative charge on B–O bonds of these borophosphate units (“mixed-former effect”) improves the mobility of lithium ions and so the ionic conductivity. For a higher boron content, the neutral borate network would be preferred with a majority of trigonal BO_3 units such as metaborate, pyroborate and orthoborate thereby reducing the connectivity between phosphate and borophosphate networks and so the ionic conductivity.

3.2. After the solder-reflow type thermal treatment

Besides an increase of the ionic conductivity with the boron content, earlier reports, as works of Carta et al. [10], Zielnok et al. [11,12], and other authors [13–16], on lithium borophosphate bulk glasses have suggested that the “mixed-former effect” of B–O–P structural units induces a better thermal stability. To study the effect of the solder-reflow type thermal treatment on the electrical properties of lithium borophosphate thin films, all samples were heated at 260 °C in a glass tube under primary vacuum to avoid any surface contamination, as explained in [Experimental procedure](#) section.

For all samples, the chemical compositions deduced from microprobe analysis and ICP before and after annealing at 260 °C are similar. In what follows, to facilitate understanding, the thermally treated samples will be named LiPOB-x-tth.

The evolution of the ionic conductivity measured at 25 °C and the activation energy for the as-deposited and thermally treated samples are respectively shown in [Figs. 8 and 9](#). From impedance spectra, there

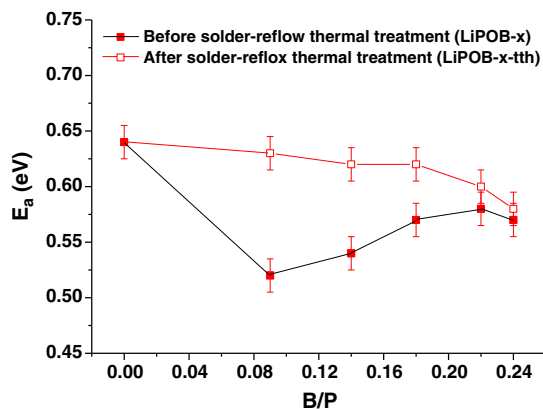


Fig. 9. Activation energy before and after solder-reflow thermal treatment as a function of B/P ratio.

is no evidence of an interface between stainless steel electrode and solid electrolyte. Whereas the ionic conductivity decreases with the boron content for the as-deposited samples, it linearly rises for the thermally treated ones. A brief annealing at 260 °C greatly enhanced the ionic conductivity by a factor of 2.4 for the LiPOB-25, from $0.8 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$ to $1.9 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$ at room temperature ([Fig. 8](#)). This evolution was previously observed for lithium and sodium borophosphate bulk glasses with similar compositions and explained by the “mixed-former effect” of borate and phosphate [10,13–17,19,24]. As the thickness and the chemical composition are not modified by the thermal treatment which takes place at a temperature lower than the glass transition temperature [17], we could expect some modifications of the local structure. The XPS analyses are under progress to understand the modifications of the local structure induced by the solder-reflow and will be presented in a forthcoming paper.

For the low rate of boron, the activation energy increases slightly with the thermal treatment ([Fig. 11](#)). For a B/P ratio equal to 0.24 (LiPOB-25), the activation energy spends from 0.57 eV to 0.58 eV further to the heat treatment instead of 0.57 to 0.80 eV for the LiPON thin film presented in a previous paper [9], indicating that the significant addition of borate within the glass network allows to increase the thermal stability of the solid electrolyte and to improve its electric performances.

As a consequence, for a process of Li-ion microbatteries with a step of solder-reflow treatment, LiPOB-25 would be a potential candidate to replace LiPON. Furthermore, the synthesis of this material has an advantage of the industrial point of view. Indeed, this solid electrolyte thin film can be prepared by unreactive sputtering, guaranteeing a greater reproducibility than by reactive sputtering and a better durability of the drop target. The nitrogen gas discharge used during reactive sputtering for the process of LiPON reacts with the target and leads to poisoning the target [25]. It is difficult to ensure good reproducibility and uniformity of deposition from an industrial point of view. The increase of pre-sputtering time before deposition could maintain a reproducibility but lead to an increase of cost. In opposition, the borophosphate solid electrolyte LiPOB-25 presents better performances after solder-reflow thermal treatment and is processed under argon discharge gas (unreactive sputtering). The use of nitrogen discharge gas, during the realization of the complete stack, can lead to a surface reaction with the positive electrode deposited below changing its properties. It would be interesting to prepare different target from the glass covered to avoid the depletion of lighter element.

4. Conclusion

In order to improve the solid electrolyte behavior after a solder-reflow process, lithium borophosphate thin films were prepared by radio-frequency sputtering using five different targets having the following compositions $x \text{ LiBO}_2 \cdot (100 - x) \text{ Li}_3\text{PO}_4$ ($x = 5, 10, 15, 20, 25$) in a pure argon discharge gas.

In the absence of a thermal treatment, the addition of a small amount of boron (B/P equal to 0.1) leads to an increase of the ionic conductivity compared to a pure lithium phosphate thin film. The glass network tends to keep local structure characteristic of the phosphate network with some borophosphate units which increase the network cross-linking. The delocalization of the negative charge on B–O bonds of these borophosphate units (“mixed-former effect”) improves the mobility of lithium ions and so the ionic conductivity. However, for B/P higher than 0.12, the ionic conductivity decreases. Thorough XPS analyses have shown that local structure is slightly different than the one of the corresponding bulk glasses. The borate groups have less reacted with the phosphate groups to form a hybrid borophosphate network than that for the bulk glasses having similar composition. In this case, the borate network would be preferred with a majority of trigonal BO_3 units such as metaborate, pyroborate and orthoborate thereby reducing the connectivity between phosphate and borophosphate networks and so the ionic conductivity.

The solder-reflow type thermal treatment leads to an increase of the ionic conductivity for borophosphate glasses as well as a slight increase of the activation energy. The best ionic conductivity, $1.9 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$, is obtained after solder-reflow type thermal treatment for a B/P ratio equal to 0.24 with activation energy of 0.57 eV instead of 0.80 eV for the LiPON thin film after this solder-reflow treatment. As a consequence, for a use of Li-ion microbatteries at low temperature connected by solder-reflow process, LiPOB-25 would be a potential candidate to replace LiPON. In addition, from an industrial point of view, the sputtering of lithium borophosphate thin film under pure argon has several advantages: a better reproducibility and a better durability of the target due to an unreactive sputtering process.

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